DARBY HUYE

Phone: (225) 572-7314 — darbyhuye.com
darbyhuye AT gmail.com — Boston, MA

PROFESSIONAL SUMMARY

Systems researcher and engineer with a PhD in Computer Science specializing in observability, distributed tracing, and performance debugging for microservice architectures. Experienced in developing low-overhead methods to improve trace data quality under data loss, and in translating complex systems data into actionable insights through visualization and data science. Research experience includes work at Meta and analysis of Alibaba trace datasets, with a focus on bridging research and practical tools for scalable, observable distributed systems.

TECHNICAL EXPERTISE

- **Observability:** Distributed Tracing (OpenTelemetry), Performance Debugging
- **Cloud & Infrastructure:** Docker, Kubernetes, Microservice Architectures, Distributed Systems
- **Languages/Tools**: Python, C/C++, SQL, JSON, MapReduce
- **Research & Data**: Topology Analysis, Distributed Data Processing, Observability Quality, Data Visualization

PROFESSIONAL EXPERIENCE

Meta, New York City, NY — 2022-2023
Research Intern, Mentor: Yuri Shkuro

- Characterized Meta's microservice architecture, analyzing topology and trace evolution for thousands of services and request types
- Quantified variability in request workflow structures to understand system behavior predictability
- Published findings at USENIX ATC '23, bridging the gap between theoretical tracing and industry-scale challenges

Tufts University, Medford, MA
Research Assistant, Advisor: Raja Sambasivan

- **Harnessing the power of holes in distributed traces** — 2024-2026
  - Developed low-overhead methods to accurately preserve topological and ordering information across data loss in traces
  - Built a reconstructor which extracts compact breadcrumbs from collected trace data and uses it to reconstruct accurate topologies and critical paths
- **Trace Data Integrity at Alibaba** — 2023

    - Identified two key categories of inconsistencies in Alibaba's trace data, invalidating prior assumptions about the data
    - Designed an algorithm to remedy errors, ensuring trace reconstruction prioritizes accuracy
    - Open-sourced corrected traces and evaluated the impact of different error-handling techniques used in prior research
- **Industry-Academic Understanding of Microservices Gap Analysis** 2021-2022
    - Conducted interviews with industry practitioners to analyze real-world perceptions and challenges in microservices
    - Deployed and evaluated academic microservice testbeds, identifying key design decisions and their implications
    - Highlighted mismatches between industry-scale microservices and the limited design space of academic testbeds

TECHNICAL PROJECTS & LEADERSHIP

- **Systems Mentorship (MIT Primes)**: Led high school students in CS research, directing projects on aggregate trace visualizations and performance uncertainty localization
- **Open Source Contributions**: Released corrected Alibaba traces and CASPER, a tool which identifies inconsistencies in Alibaba's trace data model and corrects them accurately
- **Teaching Assistant**: Assisted in teaching and grading for Computer Graphics, Cloud Computing, and Debugging Cloud Computing courses

EDUCATION

PhD Tufts University, Computer Science Jan 2021 - May 2026
Advisor: Raja Sambasivan

MS Tufts University, Data Science Jan 2021
Advisor: Raja Sambasivan

BS Tufts University, Computer Science & Mathematics May 2020
Advisors: Ming Chow & Misha Kilmer

SELECTED PUBLICATIONS

Zabcic-Matic, T, Huye, D, Lan, L., Sambasivan, R., "Bridges as a primitive for accommodating data loss in distributed tracing," Conditionally accepted at ACM Symposium on Cloud Computing (SoCC'26). DOI: coming soon

Astolfi, T., Silai, V., Huye, D, Lan, L., Sambasivan, R., Bater, J., "Dynamic read & write optimization with TurtleKV," accepted at VLDB 2026. DOI: coming soon

Huye, D, Lan, L., Sambasivan, R., "Systemizing and mitigating topological inconsistencies in Alibaba's microservice call-graph datasets," 2024 ACM/SPEC ICPE 2024. DOI: https://dl.acm.org/doi/10.1145/3629526.3645043

Huye, D., Shkuro, Y., Sambasivan, R., "Lifting the veil on Meta's microservice architecture: Analyses of topology and request workflows," 2023 USENIX Annual Technical Conference (USENIX ATC 23). DOI: https://www.usenix.org/system/files/atc23-huye.pdf

Toslali, M., Ates, E., Ellis, A., Zhang, Z., Huye, D Liu, L., Puterman, S., Coskun, A., Sambasivan, R., "Automating instrumentation choices for performance problems in distributed applications with VAIF," Proceedings of the 12th ACM Symposium on Cloud Computing (SoCC'21). November 1st to November 3rd, 2021. DOI: https://doi.org/10.1145/3472883.3487000